Code-Layout, Readability and Reusability

Date	27 June 2026
Team ID	
Project Name	Credit Card Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:
This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Followed PEP-8 coding standards
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, model, dataset, and application code
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Used descriptive names such as credit_df, model, prediction, applicant_data
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict(), preprocess_data(), train_model()
5	Code Comments	Inline and block comments explain logic	Yes	Added comments for preprocessing, model training, and prediction steps
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules for preprocessing, model

				training, and Flask application
7	No Redundant Code	Duplicate or unused code is removed	Yes	Optimized and cleaned the codebase
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling implemented in Flask

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python (Pandas, NumPy)	Model Training & Prediction	High
2	Machine Learning Model	Scikit-learn, XGBoost	Flask Prediction Module	High
3	Flask Web Application	Flask (Python)	User Interface & Prediction	High
4	HTML/CSS Templates	HTML, CSS, Bootstrap	Web Pages	Medium
5	Model Serialization (model.pkl)	Pickle	Real-time Prediction	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with a clear folder structure
Readability	5	Easy-to-read code with meaningful names and comments
Reusability	5	Modular design allows reuse of preprocessing and prediction components
Documentation / Comments	5	Proper comments and documentation provided throughout the project
Overall Score	5/5	Clean, maintainable, reusable, and scalable code